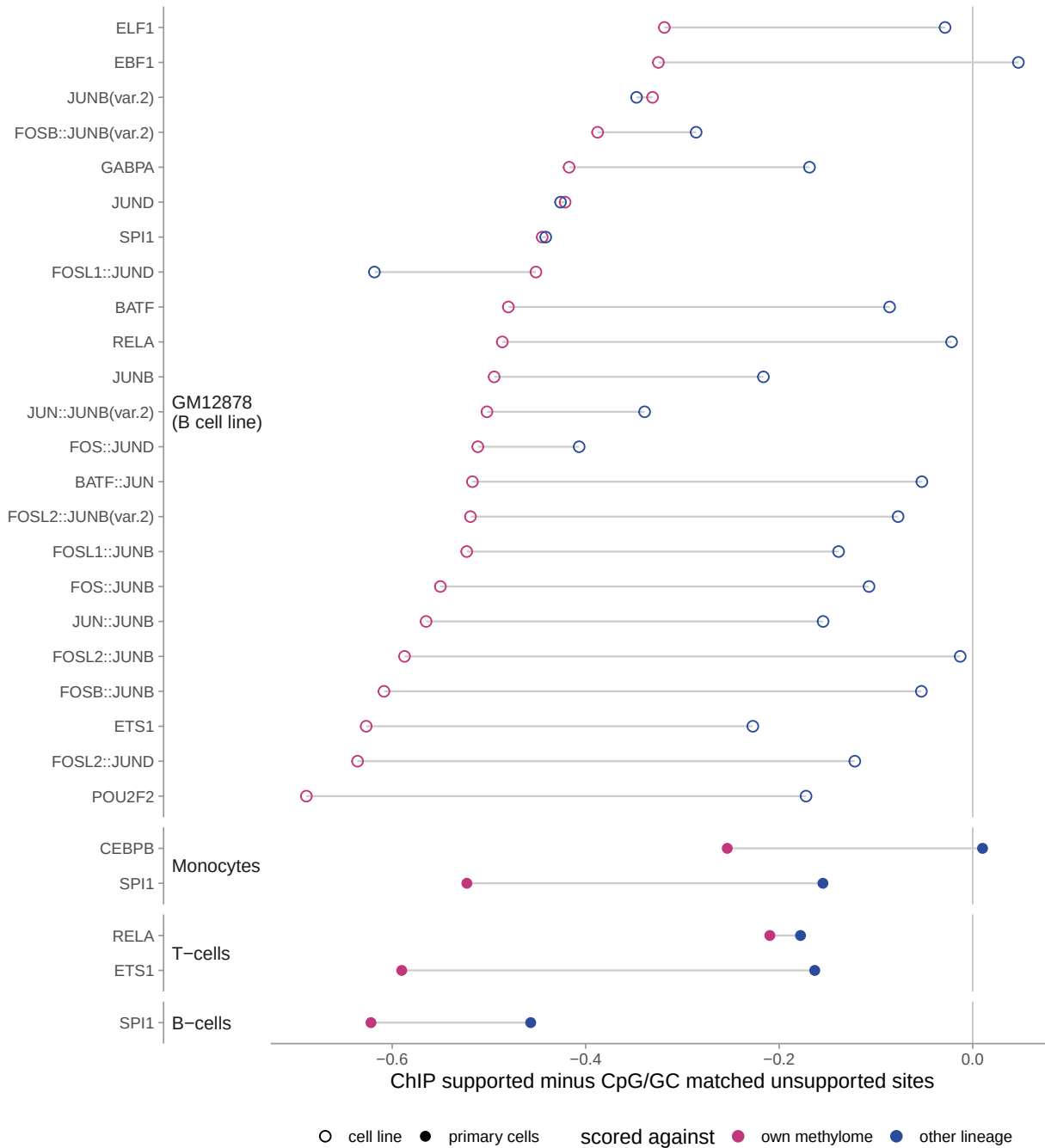


A

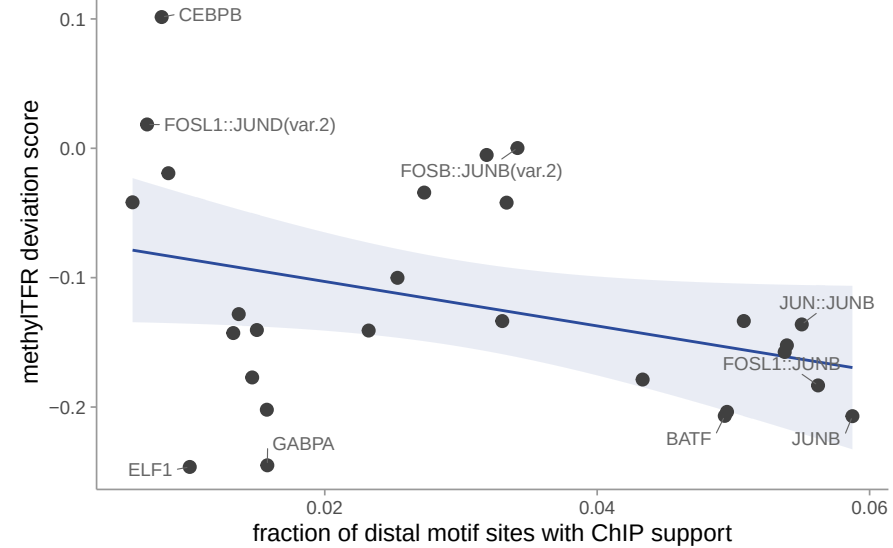
Every comparison, one row per motif and methylome

the gap between the two points is what a shared open chromatin explanation cannot produce

**B**

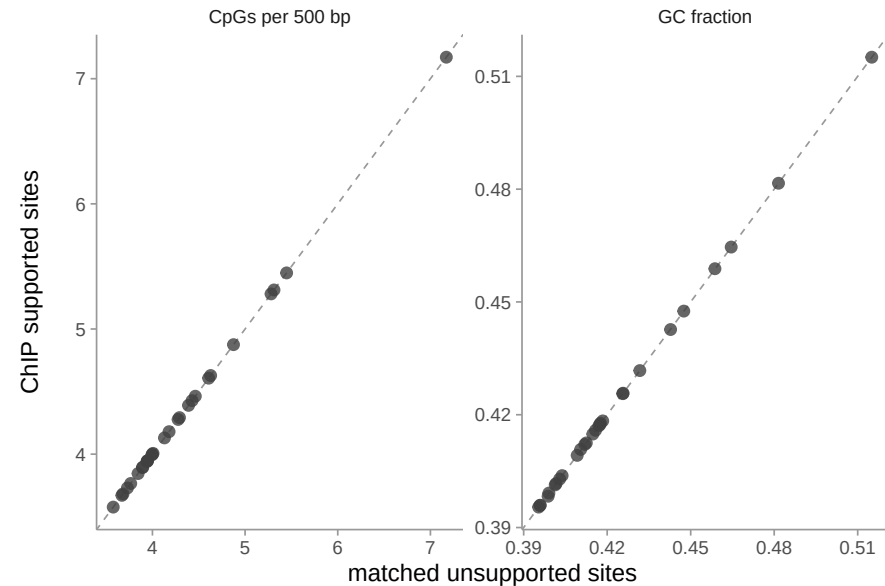
Motif level, GM12878

Spearman rho = -0.37 , $p = 0.060$, $n = 27$ motifs. Most are redundant AP-1 dimers so n overstates the evidence. Extremes labelled.

**C**

Covariate balance after matching

points on the diagonal mean the two groups are compositional twins



Peaks: ReMap2022, UniBind and ENCODE, hg38, consensus of ≥ 2 datasets where available, trimmed to ± 250 bp.
 Sites: methylTFRAnnotationHg38 jasper2020, distal only; supported = motif centre inside a peak.
 Background: unsupported sites matched 1:1 on CpG count (exact) and GC (± 0.02) in ± 250 bp, without replacement.
 Methylomes: ENCODE WGBS, GRCh38, coverage ≥ 5 . GM12878 shares a donor with its ChIP; the three primary methylomes are one donor (ENCODE661BYS).
 GM12878 supplies breadth across factors, the primary cells supply relevance. K562 was excluded as a leukaemia line with a distorted methylome.
 Expected profiles are rebuilt per site subset, since the stored per-motif table cannot distinguish subsets of one motif.